

DEVELOPMENT OF AUTOMATIC SORTING CONVEYOR BELT USING PLC

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ABSTRACT

The automatic sorting process is used to differentiate the products manufactured in an industry to further direct it towards packaging according to height. In the paper, products are compared using diffused beam type photo-electric sensor. In the paper the system based on conveyor belt model to sort the objects which are of ideal selected height is described. The control strategy for automating the sorting process which was earlier done by manual labour involving slow, inaccurate and erroneous outputs is implemented through programmable logic controller (PLC) as a solution to all these problems. In the paper, development of ladder logic program in MicroWIN software for the PLC which operates the pneumatic actuator for sorting the objects is described. It is found that HMI and SCADA using softwares WinCC Basic in TIA Portal and WinCC Professional respectively makes the model more user friendly and efficient with controls and trends on a single screen.

Key words: PLC, SCADA, Automatic sorting, TIA Portal, Sensor

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1. INTRODUCTION

It is well understood that manufacturing industries play a vital role in the development of any country and its economy. This is why the countries which have relatively higher manufacturing rates are known to be developed whereas those with lesser manufacturing are still underdeveloped. Hence, the development of manufacturing industries is of utmost importance and since it depends upon the innovations in the manufacturing process, it is a significant sector that calls for most effective technology to be implemented. Generally the manufacturing industries keep on manufacturing similar models of products with little variation in their height, color, weight and shape. Hence, sorting plays an important role there.

In older days, it was possible to utilize manual labour for sorting the objects. But in the world of technology today where there are speed running industries, the production rate has increased tremendously. In such cases industries can't bear human errors for sorting these products that too associated with high labour costs and time taking procedures. This gives rise to the necessity to reduce manual effort and also human error by replacing the conventional methods of sorting in areas involving hectic sorting with cut-cost user friendly automated systems. HMI for automatic conveyor system is considered the primary way of controlling each operation in a user-friendly way. Automating every sector of industry is an important step towards increasing efficiency, lowering cost and avoiding human related errors. Automated machines can perform a repetitive motion consistently without involving prospects of injuries or accidents thereby avoiding higher costs. Along with offering faster rates of throughput, savings are made with the wage cutting and immediate returns in efficiency leading to a faster ROI in this investment. Sorting is one such process which is witnessing transition of human unskilled labour to automated machinery in the manufacturing industries owing to the many advantages. These sorting systems are a bridge between the production and packaging machinery system used mainly in the manufacturing and logistics industries in which products are distinguished and separated on the basis of some parameter like size, height and colour. In the paper an Automatic Sorting Conveyor Belt machine which can process automatically the objects travelling on the conveyor belt and further sort them being distinguished on the basis of height is developed. A precise value of the height parameter is set which is used as a condition to either allow the objects to pass through to next step on the conveyor or be discarded in a different space so as to be collected or feed it to a different destination. The control opted to achieve this functionality is through the programmable logic controller PLC S7-200 which is dumped with the logic for the connected components to perform the required operations. The movement of objects on the conveyor belt is done by using motor and the objects are taken to the sensing part where the photo-electric sensor senses the height of the object and gives its signal to the PLC which takes a decision based on that and further controls the actuator to either reject the object by the push of piston or allow it to go further. The automatic sorting conveyor belt machine is a very practical example of automation in the manufacturing and processing industries. Furthermore, also equipping it with HMI and SCADA will expand its capabilities and reliability by making it more user-friendly. In the paper the automation of sorting method is depicted which can be further collaborated with many more techniques to be utilized in large scale industry manufacturing for a fully functional automated unit. The paper also described development of ladder logic program in MicroWIN software for the PLC which operates the pneumatic actuator for sorting the objects.

2. PROPOSED METHODOLOGY

After the manufacturing process, the job pieces of different sizes move on the same conveyor belt which need to be further sorted for packaging or shipment purposes. As soon as the green start button is pressed, the conveyor belt should start moving in the forward direction (identifying the suitable motor for this application) carrying the job pieces which are put on it manually towards the photoelectric sensor which gives the signal if the object is not of suitable height. The output is processed in the PLC program which controls the actuator to either allow the object to pass through it or discard it with the piston push. This process should run in a repetitive cycle until the stop red push button is pressed to stop the conveyor belt. The methodology adopted for the implementation of this working functionality is stated as shown in the fig1.

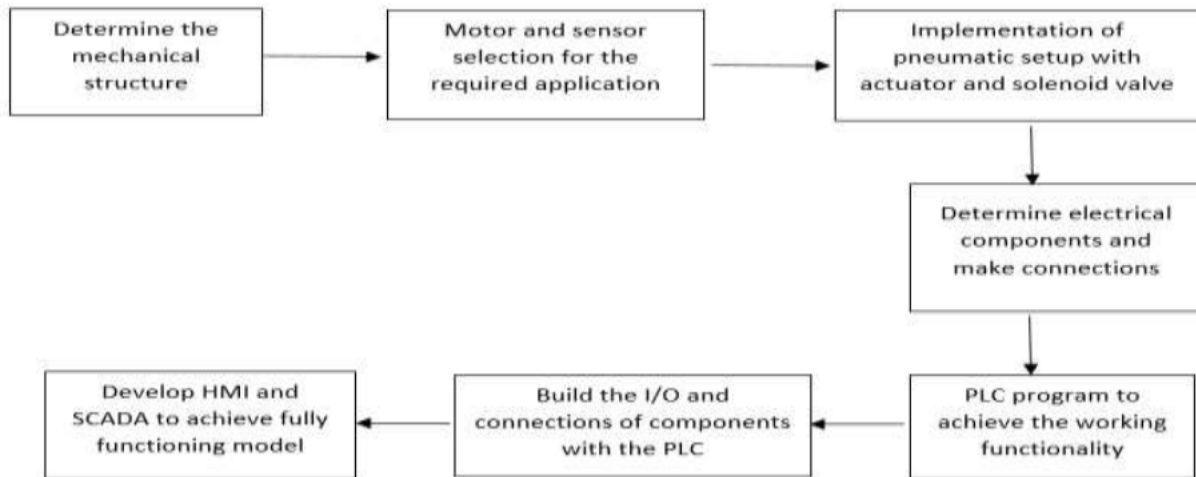


Figure 1 Block Diagram of PLC based Automatic Sorting Conveyor Belt System

2.1. Development of Hardware

Setup of the hardware model with all the required electrical, electronic, pneumatic and mechanical components is achieved which was the first objective towards the accomplishment of this project of automatic sorting conveyor belt machine. According to work flow, motor is selected for the direction of movement with relays. Controls given through push buttons and working is controlled by the program dumped in PLC which reads the photo-electric sensor signal and gives the output through actuator which is single acting cylinder with a spring reload. MCB and power supply power the circuit for its working. All these components with their specifications are listed in the Table1.

Table 1 List of Components

S. No.	Component Name	Type	Specification
1	Conveyor Belt	PVC	192cm
2	DC Motor	Oriental Motor BLH Series	BLDC 15W
3	MCB	Schneider	6A DP
4	SMPS	Siemens	24VDC/5A
5	Photo-Sensor	Diffuse Reflective	10V-30V
6	5/2 way solenoid valve	5/2type single acting	1/4inch
7	Single Acting cylinder	C85 series	50mm stroke
8	PLC	Siemens S7-200	224XPDC/DC/DC

2.2. Mathematical Analysis & Calculation

Motor-Torque Calculations

The conveyor belt travel length = 96cm = 0.96m

Radius of pulley= Radius of motor rotor = 0.049m

Radius of the roller(r) = 1.75cm= 0.0175m

Coefficient of friction between the pulley and conveyor is very low and hence can be neglected.

The requirement of this project is such that the conveyor belt must be able to bear travelling an approximate weight of 20 kg in 4-5 seconds

Required speed of conveyor belt= $96/4.5 = 21.33\text{cm/s}$

Hence, Linear speed of roller = 21.33cm/s

$$= 21.33 \times 60 / 2 \times \pi \times r$$

$$= 116\text{rpm} \approx 120\text{rpm}$$

Hence it is required a motor of about 120rpm for the required implementations.

Also, Load Torque required for running a belt conveyor:

$$\text{Torque (T)} = F \times r$$

$$= m \times (v/t) \times r$$

$$= (20\text{kg} \times (0.22\text{m/s}) \times 0.0175\text{m}) / 0.5$$

$$= 0.154 \text{ N-m}$$

Hence a motor with rated torque of 0.23Nm is sufficient to rotate the allowable range of weight on the conveyor belt. Therefore, a motor of rated torque of 0.23N-m is chosen with a variable speed range of 20-600rpm and 5:1 gear box providing 120rpm.

2.3. Electrical Drawing

An electrical drawing is a type of technical drawing that shows information of connections of all the components in an industrial project. Any electrical working drawing consists of lines, symbols for components and various notations to accurately convey the engineer's design to the workers who has to install the electrical system on the location. The Power Line connections, Relay Connections and PLC Connections have been made which are shown in the Figures 2-4.

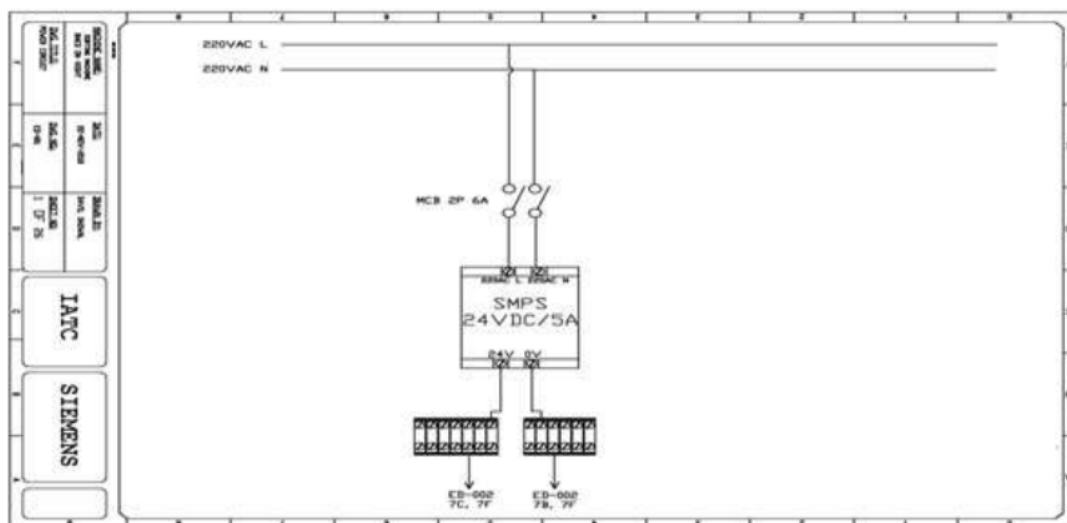


Figure 2 Power Line Diagram

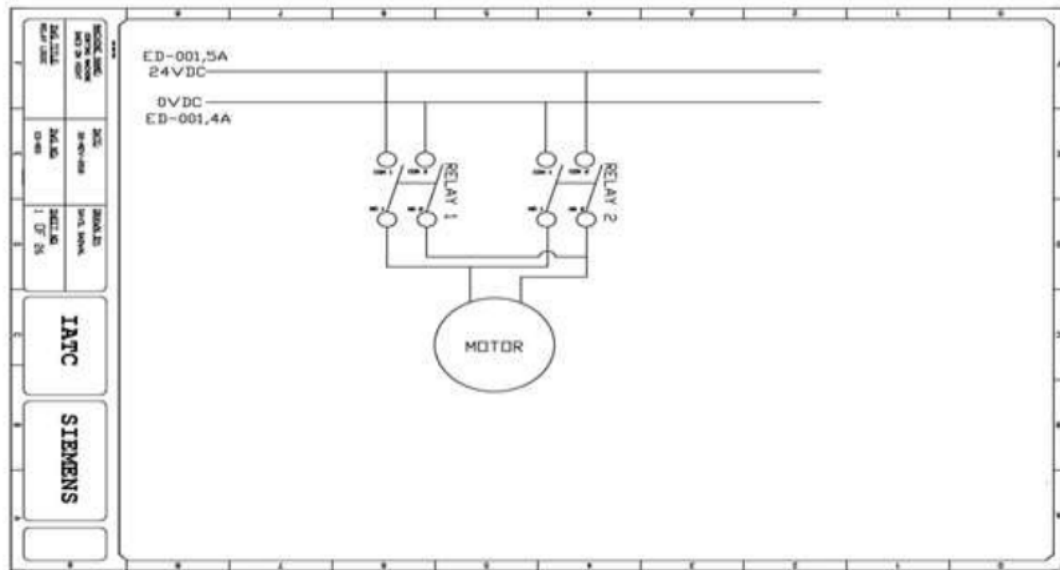


Figure 3 Relay Connection Diagram

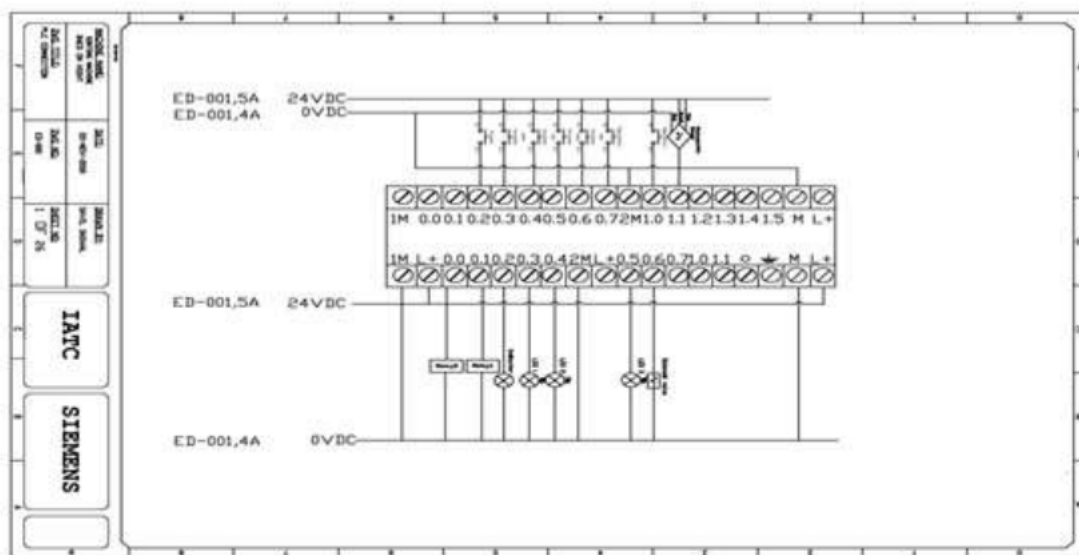


Figure 4 PLC Connections

3. HMI & SCADA IMPLEMENTATION

The program of MicroWIN V14 for PLC S7-200 replicated for the HMI using WinCC Flexible for TP 177B HMI device and communication setup established via Profinet and the graphics are designed by picking and dropping from the library and giving them appearance settings and visibility controls so that the movement can be shown as depicted in figure 5.

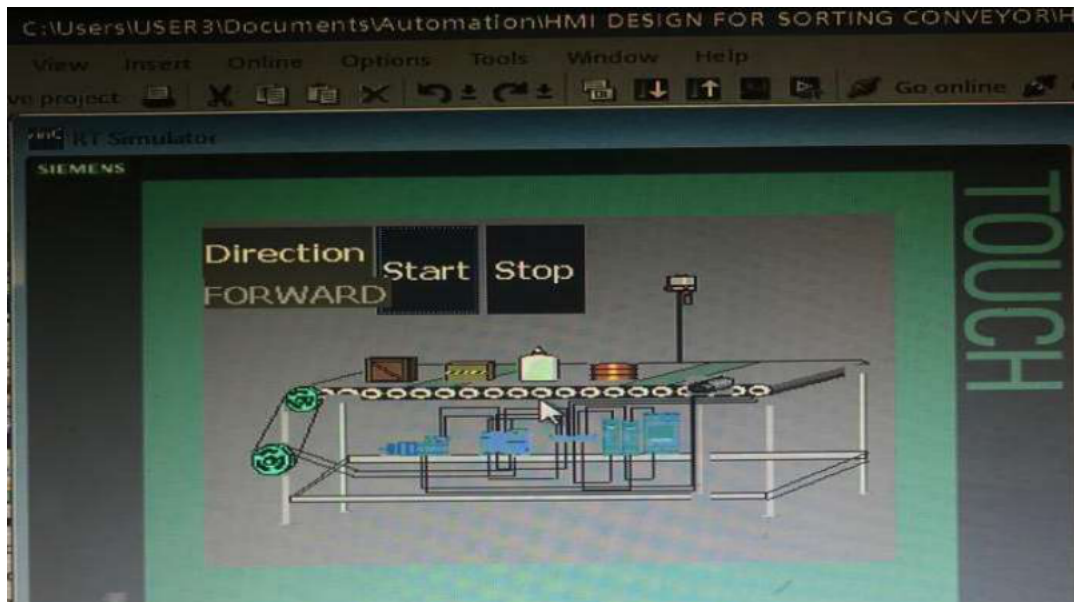


Figure 5 HMI Root Screen

Since SCADA helps in monitoring, control and data acquisition, SCADA screens are designed for monitoring and control and then given on a single screen. Total of five SCADA screens designed as below:

Screen1: HMI View which has the same graphical design as the HMI Screen and hence gives the opportunity to monitor the operation directly.

Screen2: Alarm View which helps to show warning when the total number of faulty objects reaches a limit, here set 10, and for the project completion which means the total number of objects sorted reach a limit, here set 20. Therefore, for this two outputs are activated high using counters and the program is as shown in Figures 6-7.

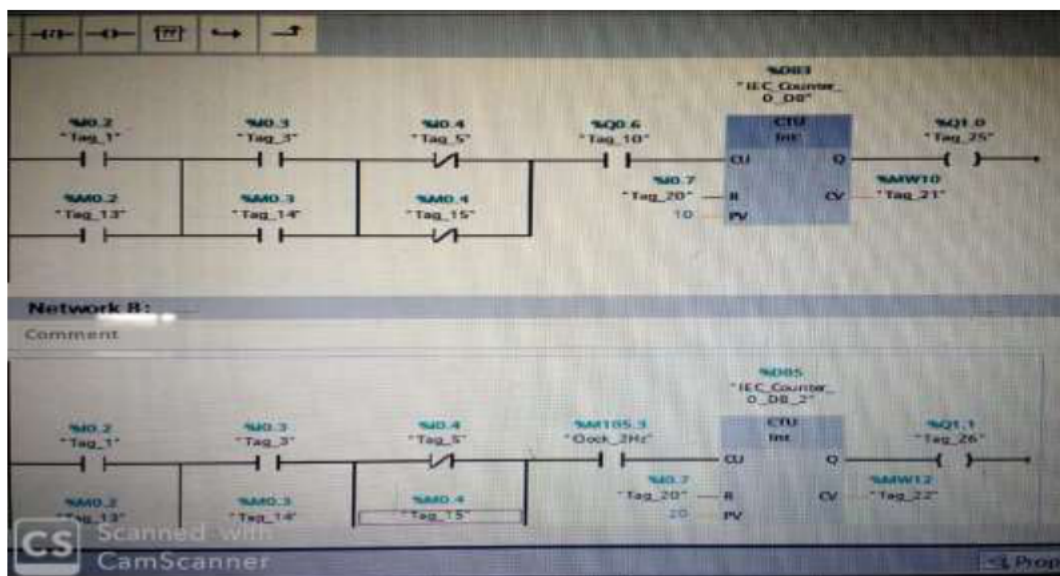


Figure 6 SCADA Program I

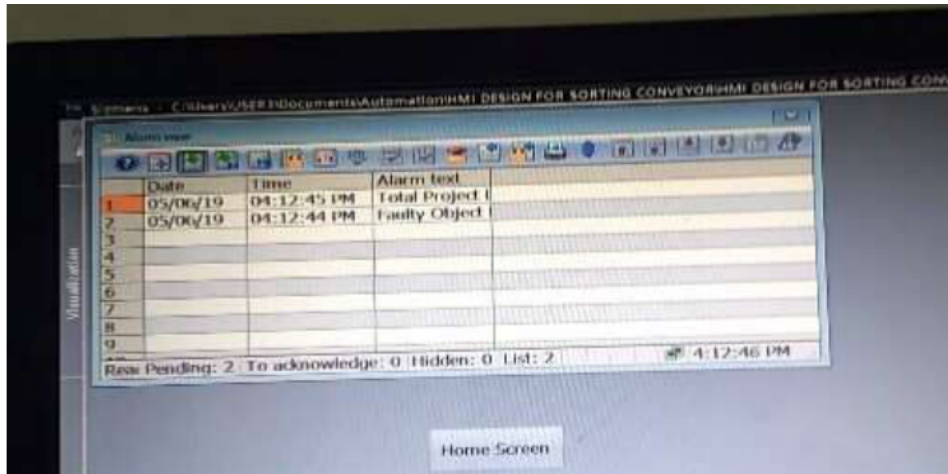


Figure 7 Alarm View

Trend View as presented in figure 8 is a graphical representation of the values of parameters of faulty, correct and total objects being sorted which are stored as memory addresses in counters activated by the output of the actuator and the correct objects quantified by the mathematical calculation of subtraction from the total objects .

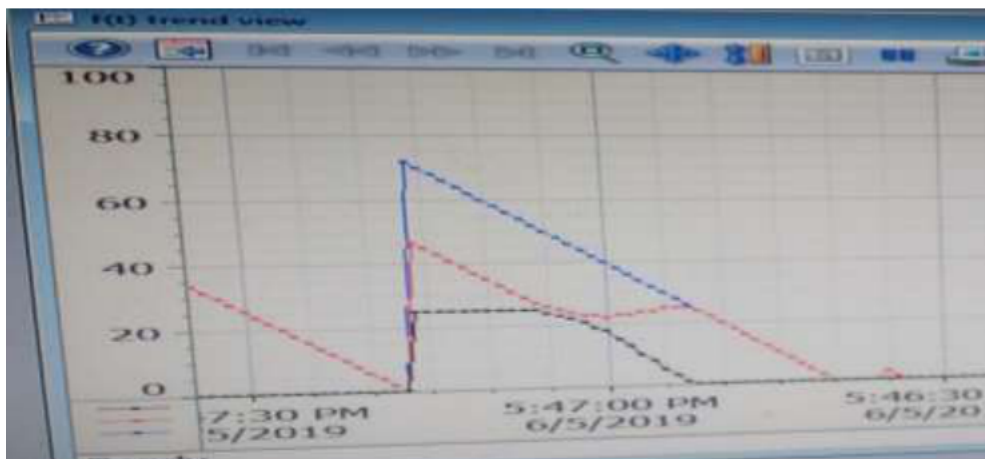


Figure 8 Trend View

4. RESULTS AND DISCUSSION

4.1. Hardware Implementation

For the accomplishment of the proposed work, the hardware has been tested for the relay connections for motor rotation to run the conveyor belt forward and backward. Also, the final model tested for the complete functionality through the controls panel with push buttons and toggle switch. Two Double Pole Double Throw Relays switched using green and red push buttons for start and stop respectively such that at one time only one relay operates which is activated by another mode selector switch for direction forward or reverse. The start push button starts the Relay 1 for forward and even if the mode is changed to reverse, green push button does not switch the Relay 2 as shown in figure 10.



Figure 10 Switching of Relays

In this figure 11 Conveyor Belt Sorting Machine setup finally tested with all controls and the setup is able to sort objects of height 6cm and discards all the other objects with the piston. Also the start stop and direction control function is tested which gives quick operation.

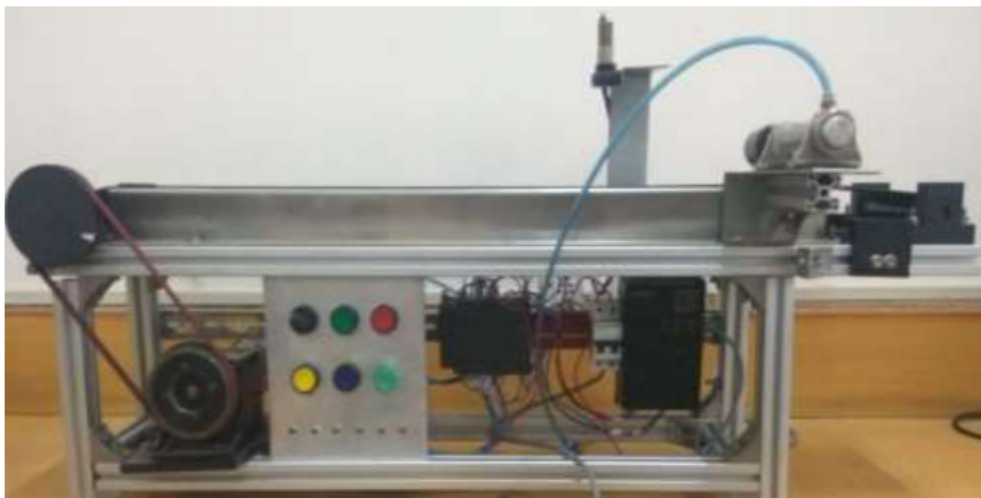


Figure 11 Conveyor Belt Sorting Machine

4.2. Software Simulation

The graphics and controls designed on HMI were tested as can be seen in Figure 12 HMI Forward and Reverse Direction, the conveyor doesn't sort objects in the reverse direction while only runs in the forward direction when the Start green push button is pressed.

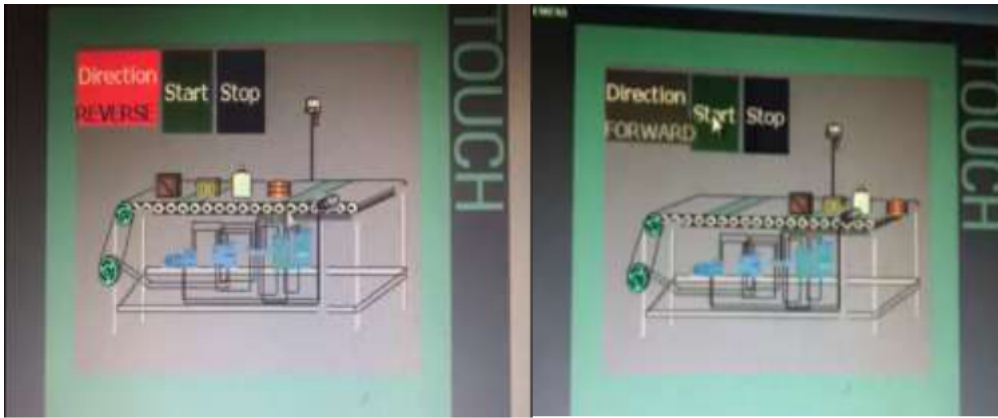


Figure 12 HMI- Forward and Reverse Direction

SCADA Screen simulated as shown in Figure 13 depicts that all four screens can be monitored on a single screen with trends, tables and alarms.

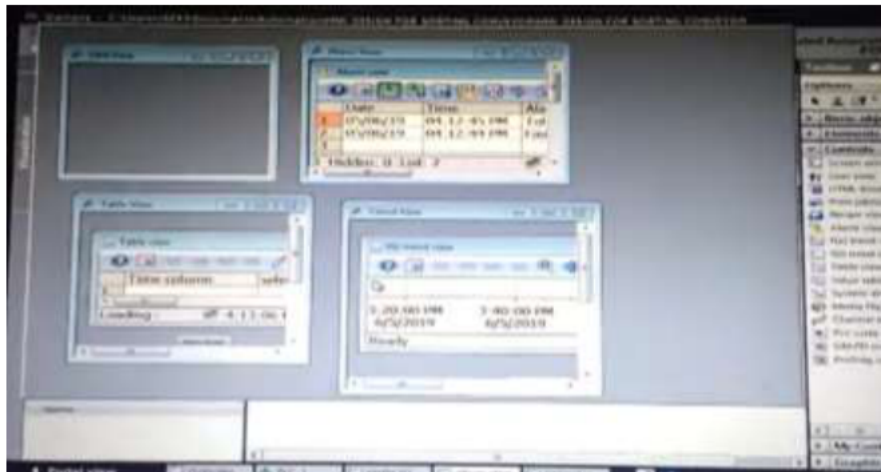


Figure 13 SCADA Simulation

5. CONCLUSION

In the paper Oriental Motor 15W DC Brushless motor has been selected according to the speed and torque requirements (120rpm, 0.23 N-m) in the application of Automatic Sorting Conveyor Belt which has a parallel shaft and 5:1 gear box attached to it. Also Line side components connection has been built with MCB and SMPS terminating to the terminal blocks of 0V and 24V where the connections of relay through motor and pneumatic control connections meet. PLC Program for the complete functionality of rotating the motor in both forward and reverse directions, sensor and actuator has been developed and achieved by using I/P I0.2 as a toggle switch, I0.3 as a start push button (green), I0.4 as a stop push button (red), I0.5 as the sensor I/P and Q0.6 as the actuator O/P. It can be concluded that HMI Screen designed with all the graphics and controls and the movements and visibility given for operation and SCADA system implemented on a single screen which has window controls of four screens for monitoring, trends and alarms may be used in manufacturing industries to improve quality of products.

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